

A candidate counterexample to Melnikov's valency-variety problem

github.com/graph-theory-AI — machine-generated candidate writeup

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Abstract

A 37-vertex graph has chromatic number 3 and degree set $\{0,1,\dots,29\}$, making the proposed right-hand side equal to 3; moreover, 37 vertices is minimum. This candidate disproof was generated by GPT-6 Astra in a single model pass for the Graph Theory AI project. It was selected from the campaign because the model marked it `would_publish: true`; it has not been checked by the adversarial referee, a human mathematician, or a novelty search.

1 The problem and the claimed result

This document concerns [Melnikov's valency-variety problem](http://www.openproblemgarden.org/op/melnikovs_valency_variety_problem) from http://www.openproblemgarden.org/op/melnikovs_valency_variety_problem.

1.1 Problem statement

Problem The valency-variety $w(G)$ of a graph G is the number of different degrees in G . Is the chromatic number of any graph G with at least two vertices greater than

$$\left\lceil \frac{\lfloor w(G)/2 \rfloor}{|V(G)| - w(G)} \right\rceil ?$$

1.2 Catalog context

According to Jensen and Toft [JT, p. 90], the problem is due to Melnikov and was mentioned by Vizing [V] and Zykov [Z]. According to Zykov [Z], Melnikov showed that the suggested lower bound would be best possible. A best possible upper bound on the chromatic number in terms of $|V(G)|$ and $w(G)$ is $|V(G)| - \lfloor w(G)/2 \rfloor$ as proved by Nettleton [N] and Dirac [D].

1.3 Claimed result

Model verdict: disproved (confidence: high).

A 37-vertex graph has chromatic number 3 and degree set $\{0,1,\dots,29\}$, making the proposed right-hand side equal to 3; moreover, 37 vertices is minimum.

The model's own caveat: The counterexample has one isolated vertex, which the statement permits; bibliographic novelty has not been independently checked.

2 Model writeup

The text below is the model output verbatim, apart from moving its machine-readable verdict block into the summary above and shifting Markdown heading levels for this document. Mechanical TeX defects and equation tags were normalized where needed for compilation.

2.1 A smallest counterexample

Graphs below are finite and simple, and degree 0 counts toward the valency-variety.

2.1.1 1. Explicit construction

Take disjoint vertex sets

$$\begin{aligned} A &= \{a_1, \dots, a_5\}, & B &= \{b_0, \dots, b_5\}, \\ C &= \{c_0, \dots, c_6\}, & T &= \{t_1, \dots, t_6\}, \\ U &= \{u_1, \dots, u_{12}\}, \end{aligned}$$

together with one vertex z . There are

$$5 + 6 + 7 + 6 + 12 + 1 = 37$$

vertices.

Define the edges as follows, with **no edges other than those specified**.

1. On $A \cup B \cup C$, put the complete tripartite graph with parts A, B, C .
2. Join every vertex of T to every vertex of $B \cup C$.
3. Join each of $u_7, \dots, u_{12}$ to every vertex of B .
4. Put the following edges between U and A :

$$\begin{aligned} N(u_7) \cap A &= \{a_5\}, \\ N(u_8) \cap A &= \{a_4, a_5\}, \\ N(u_9) \cap A &= \{a_3, a_4, a_5\}. \end{aligned}$$

5. For $i \in \{10, 11, 12\}$, join u_i to c_j precisely when

$$j \geq 13 - i.$$

Thus these three neighborhoods in C have sizes 4, 5, 6.

6. For $1 \leq i \leq 6$, join u_i to b_j precisely when

$$i + j \geq 7.$$

Also put the edges

$$u_1c_4, \quad u_2c_5, \quad u_3c_5, \quad u_4c_6, \quad u_5c_6, \quad u_6c_6.$$

In particular, $T \cup U$ is independent, and z is isolated.

2.1.2 2. Verification

2.1.2.1 Degrees The complete degree table is

Vertices	Degree	Number of vertices
z	0	1
$u_i, 1 \leq i \leq 12$	i	12
$a_1, a_2, t_1, \dots, t_6$	13	8
a_3, a_4, a_5	14, 15, 16	3
$c_j, 0 \leq j \leq 6$	$17 + j$	7
$b_j, 0 \leq j \leq 5$	$24 + j$	6

Here are direct checks of the less immediate entries.

- Each a_i has 13 neighbors in $B \cup C$, and the extra degrees supplied by U are 0, 0, 1, 2, 3.
- Each b_j has 12 neighbors in $A \cup C$, six in T , six among $u_7, \dots, u_{12}$, and exactly j among $u_1, \dots, u_6$.
Thus

$$d(b_j) = 12 + 6 + 6 + j = 24 + j.$$

- Each c_j has 11 neighbors in $A \cup B$, six in T , and exactly j in U . Hence

$$d(c_j) = 17 + j.$$

- For $1 \leq i \leq 6$, the vertex u_i has $i - 1$ neighbors in B and one in C . The definitions likewise give $d(u_i) = i$ for $7 \leq i \leq 12$.

Consequently,

$$\{d(v) : v \in V(G)\} = \{0, 1, \dots, 29\},$$

so

$$w(G) = 30.$$

2.1.2.2 Chromatic number The following are three independent sets partitioning $V(G)$:

$$I_1 = A \cup T \cup \{u_1, \dots, u_6, u_{10}, u_{11}, u_{12}, z\},$$

$$I_2 = B,$$

$$I_3 = C \cup \{u_7, u_8, u_9\}.$$

Therefore $\chi(G) \leq 3$. Since $a_1 b_0 c_0 a_1$ is a triangle,

$$\chi(G) = 3.$$

2.1.2.3 Failure of the proposed inequality For this graph,

$$|V(G)| - w(G) = 37 - 30 = 7$$

and

$$\left\lceil \frac{\lfloor w(G)/2 \rfloor}{|V(G)| - w(G)} \right\rceil = \left\lceil \frac{15}{7} \right\rceil = 3.$$

Thus the proposed conclusion is $3 > 3$, which is false.

The graph has exactly one isolated vertex; its other 36 vertices form a connected component.

2.2 3. Minimality: no counterexample has fewer than 37 vertices

The following argument is not needed for the disproof, but establishes that the order of the construction is optimal.

Write

$$n = |V(G)|, \quad k = \chi(G), \quad t = n - w(G).$$

For a simple graph on at least two vertices, $w(G) \leq n - 1$, so $t \geq 1$.

The proposed inequality is equivalent to

$$\left\lfloor \frac{w(G)}{2} \right\rfloor \leq (k - 1)t,$$

and hence to

$$\boxed{n \leq (2k - 1)t + 1.} \quad (1)$$

We prove two facts:

1. every bipartite graph satisfies (1);
2. every graph with $t \leq 6$ satisfies (1).

These imply minimality immediately. Indeed, a counterexample has $k \geq 3$ and

$$n \geq (2k - 1)t + 2 \geq 5t + 2.$$

If $n \leq 36$, this forces $t \leq 6$, contradicting the second fact.

2.2.1 3.1 Bipartite graphs

First suppose the graph has no isolated vertices. Take a bipartition with part sizes $a \leq b$.

All degrees lie in $\{1, \dots, b\}$, giving $w \leq b$. The larger part contributes at most a degree values, and the smaller part has only a vertices, so $w \leq 2a$. Therefore

$$n = a + b \geq \frac{w}{2} + w,$$

or equivalently

$$n \leq 3(n - w) = 3t. \quad (2)$$

If a nonempty graph has $r \geq 1$ isolated vertices, deleting all of them gives a graph G_0 with

$$n_0 = n - r, \quad w_0 = w - 1, \quad t_0 = t - r + 1.$$

Applying (2),

$$n = n_0 + r \leq 3t_0 + r = 3t - 2r + 3 \leq 3t + 1.$$

Edgeless graphs satisfy the original inequality directly. Thus every bipartite graph satisfies (1).

2.2.2 3.2 A degree-spread estimate

Consider a graph of even order $2m$, with its degrees arranged as

$$d_1 \leq \dots \leq d_{2m}.$$

Let L be the first m vertices and H the last m . Put

$$b = d_m, \quad \delta = d_1, \quad D = \sum_{v \in H} d(v) - \sum_{v \in L} d(v).$$

Then

$$D = \sum_{i=1}^{2m} |d_i - b| = 2e(H) - 2e(L). \quad (3)$$

In particular, D is even.

If there are w distinct degrees, selecting one occurrence of each degree in the absolute-value sum gives

$$D \geq D_0 := \left\lfloor \frac{w^2}{4} \right\rfloor. \quad (4)$$

Indeed, the minimum sum of distances from an integer b to w distinct integers is $\lfloor w^2/4 \rfloor$.

We also need the following refinements, using that every degree is at least δ :

- If $w = 2r + 1$ and $b \leq \delta + r$, then

$$D \geq r(r + 1) + (\delta + r - b)^2. \quad (5)$$

- If $w = 2r$ and $b \leq \delta + r - 1$, then

$$D \geq r^2 + (\delta + r - 1 - b)(\delta + r - b). \quad (6)$$

For completeness, under the hypothesis of (5), the $2r + 1$ closest distinct integers at least δ are

$$\delta, \delta + 1, \dots, \delta + 2r.$$

Their distances from b sum to the right-hand side of (5). The same argument with $\delta, \dots, \delta + 2r - 1$ proves (6).

2.2.3 3.3 Incorporating a coloring

Fix a proper k -coloring and let h_i be the number of vertices of color i in H . Set

$$a = \min_i h_i, \quad e = e(L).$$

Since there are no edges within a color class,

$$D + 2e + \sum_{i=1}^k h_i^2 \leq m^2. \quad (7)$$

Also, any vertex of L has at most $m - a$ neighbors in H , and at most e neighbors in L . Hence

$$b \leq m - a + e. \quad (8)$$

Let

$$Q_k(m) = \min \left\{ \sum_{i=1}^k x_i^2 : x_i \in \mathbb{Z}_{\geq 0}, \sum_i x_i = m \right\}.$$

The minimum occurs when the x_i 's differ by at most one. This follows by transferring one unit from a larger entry to a smaller entry whenever they differ by at least two.

Since $w = 2m - t$,

$$D_0 = m^2 - tm + \left\lfloor \frac{t^2}{4} \right\rfloor.$$

Equations (4) and (7), together with Cauchy–Schwarz, give

$$\frac{m^2}{k} \leq Q_k(m) \leq tm - \left\lfloor \frac{t^2}{4} \right\rfloor. \quad (9)$$

For $k \geq 2$, the quadratic inequality (9) yields

t	Upper bound on m	
1	k	
2	$2k - 1$	
3	$3k - 1$	
4	$4k - 2$	
5	$5k - 2$	
6	$6k - 2$	(10)

For example, at one more than these bounds, the left side minus the right side of (9) is respectively

$$1 + \frac{1}{k}, \quad 1, \quad 2, \quad \frac{1}{k}, \quad 1 + \frac{1}{k}, \quad 3 + \frac{1}{k},$$

and the quadratic is increasing thereafter.

Only four boundary cases need refinement. Define

$$X = D - D_0, \quad E = \sum_i h_i^2 - Q_k(m).$$

Both are nonnegative integers, and E is even, because $x^2 \equiv x \pmod{2}$. From (7),

$$X + 2e + E \leq m^2 - D_0 - Q_k(m). \quad (11)$$

The relevant boundary data are

t	m	$Q_k(m)$	$m^2 - D_0 - Q_k(m)$	a when $E = 0$
1	k	k	0	1
3	$3k - 1$	$9k - 5$	0	2
5	$5k - 2$	$25k - 18$	2	4
6	$6k - 2$	$36k - 22$	1	5

(12)

2.2.3.1 The boundaries $t = 1, 3$ Assume there are no isolated vertices, so $\delta \geq 1$.

Equation (11) forces $X = e = E = 0$. For $t = 1$, write $w = 2r + 1$ with $r = m - 1$; for $t = 3$, use $r = m - 2$. In both cases, (8) and (12) give $b \leq r$. But (5), with $X = 0$, requires

$$b \geq \delta + r \geq r + 1,$$

a contradiction.

Thus these boundary orders cannot occur without isolated vertices.

2.2.3.2 The boundary $t = 5$ Again assume $\delta \geq 1$. Here

$$m = 5k - 2, \quad w = 2r + 1, \quad r = m - 3,$$

and

$$X + 2e + E \leq 2.$$

Since $D_0 = r(r + 1)$ and D are even, X is even.

We use the observation that $E \leq 2$ forces $a \geq 3$. Otherwise, some $h_i \leq 2$; because their sum is $5k - 2$, another entry is at least 6. Transferring one unit between these entries decreases the sum of squares by at least 6, implying $E \geq 6$, a contradiction.

If $X = 0$, there are two possibilities:

- $E = 0$, so $a = 4$ and $e \leq 1$;
- $E = 2$, so $e = 0$ and $a \geq 3$.

In either case, (8) gives $b \leq m - 3$. But (5), with $X = 0$, requires

$$b \geq \delta + r \geq m - 2.$$

If $X = 2$, then $e = E = 0$, so $a = 4$ and $b \leq m - 4$. Hence

$$\delta + r - b \geq 2,$$

and (5) gives $X \geq 4$, again a contradiction.

Thus this boundary also cannot occur without isolated vertices.

2.2.3.3 The boundary $t = 6$ Here no assumption about isolated vertices is needed. We have

$$m = 6k - 2, \quad w = 2r, \quad r = m - 3 = 6k - 5.$$

Thus $D_0 = r^2$ is odd. Since D is even, X is a positive odd integer. Equation (11) therefore forces

$$X = 1, \quad e = E = 0.$$

By (12), $a = 5$, and consequently

$$b \leq m - 5 = r - 2.$$

As $\delta \geq 0$, inequality (6) gives

$$D \geq r^2 + 2,$$

contradicting $X = 1$.

This boundary is impossible for every graph.

2.2.4 3.4 Completion for $t \leq 6$

The preceding estimates show that for every even-order nonempty graph with $t \leq 6$,

$$n \leq \begin{cases} (2k - 1)t, & t \text{ even,} \\ (2k - 1)t + 1, & t \text{ odd,} \end{cases} \quad (13)$$

and in the odd- t case equality in (13) requires an isolated vertex.

It follows that every graph **without isolated vertices** and with $t \leq 6$ satisfies

$$n \leq (2k - 1)t. \quad (14)$$

For even order this is already proved. For odd order, add one isolated vertex. This increases both n and w by one, leaving t and k unchanged. Applying (13) to the resulting even-order graph gives (14).

Finally, suppose a nonempty graph has $r \geq 1$ isolated vertices. Deleting them gives

$$t_0 = t - r + 1 \leq 6.$$

Applying (14) to the remaining graph,

$$\begin{aligned} n &\leq (2k - 1)t_0 + r \\ &= (2k - 1)t - (2k - 2)r + (2k - 1) \\ &\leq (2k - 1)t + 1. \end{aligned}$$

Edgeless graphs satisfy the original assertion directly. Therefore every graph with $t \leq 6$ satisfies (1).

As explained above, this excludes every counterexample on at most 36 vertices. The constructed 37-vertex graph is therefore smallest possible.

2.2.5 Conclusion and scope

The displayed conjecture is **false as stated**, and the minimum counterexample order is 37.

The isolated vertex is material: deleting it from the construction leaves a graph that satisfies the displayed inequality. Thus the construction does not by itself refute a version restricted to connected graphs. No computational search or unverified literature result is used in the argument; whether this counterexample or its minimality has previously appeared remains a bibliographic question.

A Provenance and review status

UNREFEREED MODEL OUTPUT. This document was selected solely because GPT-6 Astra labelled its own result `would_publish: true` and returned `proved` or `disproved`. It has not passed the adversarial LLM referee used for the earlier Sol campaign, has not been checked by a human mathematician, and has not been checked for novelty. Treat every mathematical and bibliographic claim as unverified.

Catalog id	melnikovs_valency_variety_problem
Catalog entry	Melnikov's valency-variety problem
Source corpus	opg
Campaign leg	OpenProblemGarden first pass (attacks_opg)
Source paper / entry	http://www.openproblemgarden.org/op/melnikovs_valency_variety_problem
Model verdict	disproved (confidence: high)
Selection criterion	would_publish: true and verdict proved or disproved
Model	gpt-6-astra, reasoning effort max, mode=pro, flex service tier
Original artifact	attacks_opg/melnikovs_valency_variety_problem/output.md
Independent review	None
Model's caveats	The counterexample has one isolated vertex, which the statement permits; bibliographic novelty has not been independently checked.

The generated Markdown and LaTeX sources for this PDF are stored under `to_review_astra/src/melnikovs_valency_variety_problem`